

BARTLESVILLE, OK, USAWMO: **723565**Lat: **36.767N**Lon: **96.017W**Elev: **218**StdP: **98.73**Time Zone: **-6.00 (NAC)**Period: **99-19**WBAN: **03959****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.7	-17.7	0.8	-9.6	-15.1	1.0	-6.8	10.3	8.3	9.0	8.8	1.8	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	37.8	24.5	36.1	24.8	34.2	24.6	26.7	33.6	26.0	33.0	25.4	32.3	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.4	30.5	24.0	19.4	29.8	23.4	18.7	29.0	84.8	33.5	81.7	33.0	79.0	32.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.3	-18.2	39.7	4.6	2.4	-21.5	41.4	-24.2	42.8	-26.9	44.1	-30.2	45.8		
(5)			-18.5	28.2	4.6	0.9	-21.8	28.9	-24.5	29.4	-27.0	29.9	-30.3	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	1.9	4.0	9.8	15.1	19.7	24.6	27.0	26.5	22.0	15.3	8.8	3.1
	DBStd	10.06	5.71	6.16	5.86	4.96	4.25	2.89	2.84	3.20	3.94	5.05	5.38	5.46
	HDD10.0	847	256	186	75	13	1	0	0	0	0	12	84	221
	HDD18.3	2234	509	403	271	121	37	1	0	0	12	121	288	472
	CDD10.0	2629	6	16	68	164	303	439	528	512	361	177	49	7
	CDD18.3	975	0	1	6	23	81	189	269	254	123	27	3	0
	CDH23.3	11148	1	10	71	278	805	2062	3245	2968	1356	322	30	1
CDH26.7	5344	0	1	13	68	257	914	1761	1624	614	91	3	0	
Wind	WSAvg	2.8	2.8	3.1	3.4	3.6	2.9	2.7	2.2	2.1	2.2	2.5	2.8	2.7
Precipitation	PrecAvg	950	37	43	79	92	120	113	76	80	107	84	68	49
	PrecMax	1489	106	174	224	245	262	382	188	273	371	193	168	145
	PrecMin	504	0	0	6	3	21	10	1	3	7	4	0	4
	PrecStd	214	28	38	54	48	61	71	52	57	75	54	52	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	24.8	28.3	31.5	32.7	36.5	39.7	40.9	37.6	32.0	26.5	21.6
		MCWB	12.2	15.1	17.9	19.6	22.6	25.2	24.3	24.3	23.7	21.8	17.1	14.8
	2%	DB	18.3	21.2	25.3	28.5	31.2	34.2	37.6	38.1	34.1	29.0	23.6	17.8
		MCWB	10.7	13.0	16.1	18.8	22.3	24.7	24.9	24.2	23.7	20.1	15.7	12.0
	5%	DB	15.4	18.0	22.7	26.6	29.6	32.8	36.0	36.3	32.4	27.1	21.2	15.3
		MCWB	9.3	11.9	14.9	17.7	21.8	24.5	25.1	24.7	23.1	18.8	14.7	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	12.2	15.1	20.4	24.4	27.8	31.5	34.1	34.2	30.4	24.7	18.4	12.6
		MCWB	7.1	9.5	13.9	17.2	21.1	23.9	25.1	24.4	22.2	18.0	12.8	8.6
	0.4%	WB	15.5	17.1	19.6	21.8	25.0	26.9	27.8	27.7	25.7	23.5	19.3	17.3
		MCDB	18.2	21.9	26.1	27.8	30.4	33.5	34.2	34.1	32.6	29.1	24.1	20.1
	2%	WB	12.2	15.0	17.9	20.6	23.8	26.1	26.8	26.6	24.7	22.2	17.5	13.9
		MCDB	16.4	19.3	22.9	26.0	29.0	32.5	33.7	33.8	31.5	26.7	21.1	16.4
Mean Daily Temperature Range	5% DB	WB	9.6	12.6	16.5	19.6	22.9	25.3	26.1	25.7	24.0	20.9	15.8	11.2
		MCDB	15.1	17.2	21.0	24.4	27.9	31.5	33.2	33.0	30.5	24.5	19.5	13.9
	5% WB	WB	7.4	9.6	14.8	18.3	21.9	24.5	25.5	25.0	23.1	19.2	13.9	8.7
		MCDB	11.7	14.7	19.5	23.1	26.6	30.3	32.5	32.2	28.8	23.1	17.8	12.3
Clear-Sky Solar Irradiance	taub	0.298	0.310	0.354	0.386	0.423	0.434	0.444	0.446	0.394	0.350	0.313	0.293	
	taud	2.485	2.465	2.378	2.313	2.269	2.267	2.284	2.241	2.363	2.467	2.508	2.515	
	Ebn at Noon	899	930	911	896	863	850	839	829	858	871	873	882	
	Edh at Noon	81	94	113	127	135	135	132	135	113	93	79	74	
All-Sky Solar Radiation	RadAvg	2.52	3.16	4.14	5.20	5.77	6.61	6.59	5.94	4.97	3.70	2.70	2.14	
	RadStd	0.24	0.48	0.25	0.34	0.54	0.58	0.40	0.39	0.40	0.44	0.31	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page